



Weld claddings of Hastelloy C276 using the GTAW cold wire feeding process

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ABSTRACT

The primary goal of this study was to analyse the microstructural and microchemical characteristics of dissimilar Hastelloy C276 weld claddings on ASTM A516 Gr. 60 steel. These claddings were produced using the GTAW cold wire feed process with varying heat input parameters. We utilised several analytical techniques, including scanning electron microscopy, transmission electron microscopy, X-ray energy-dispersive spectroscopy, and X-ray diffraction, for our microstructural and microchemical analyses. Our results demonstrated that adjusting the heat input and wire feed speed reduced the dilution level of the Hastelloy C276 weld cladding from approximately 18% to approximately 8%. This low dilution level, achieved by a single deposition layer, offers significant productivity benefits for the industry. Despite rapid cooling, microsegregation occurred, precipitating secondary phases, notably Mo-rich P -, μ - and σ -topologically close-packed phases in the interdendritic region. Pitting corrosion resistance was evaluated in accordance with ASTM G48 (Method C), showing a critical pitting temperature (CPT) of 70–75 °C for a medium dilution sample, while the lowest dilution sample showed a CPT of 65–70 °C. These findings have practical implications for industry, particularly for process optimisation and quality control.

1. Introduction

Pipelines, pressure vessels, valves, and other components at offshore installations and refineries are vital to the processing and transportation of crude oil and its derivatives. Some corrosive agents such as carbonic acid (H_2CO_3), hydrogen sulphide (H_2S), naphthenic acids ($C_nH_{2n} + ZO_2$), production water, and mineral salts (halides) [1]. These compounds make petroleum an extremely aggressive medium [2–4]. As a result, corrosion-resistant alloys, such as stainless steel and Ni-based alloys, are often used to fabricate pipes and equipment [5–9]. Advances in alloy metallurgy enabled the development of Hastelloy C276 in the 1960s, a second-generation solid-solution-strengthening alloy [10–13]. Hastelloy C276 contains a high concentration of refractory alloying elements, including chromium, molybdenum, and tungsten, which confer additional strength and excellent corrosion resistance [11, 13]. This alloy has been successfully utilised in the chemical processing industry for applications such as heat exchangers, piping, reactors,

evaporators, and columns that handle a diverse range of chemicals, including strong acids, due to its high resistance to HCl attacks [13–15]. In the petrochemical and oil & gas industries, it has been used in components that operate in corrosive streams and at high temperatures, owing to the alloy's durability [16,17]. It has also found applications in various air pollution control systems, such as flue gas desulphurisation (FGD) systems, including chimney liners, ducts, scrubbers, and fan components, owing to its resistance to corrosive flue gases [18–20].

However, the high cost of these CRAs has motivated investment in the development and optimisation of manufacturing processes for special materials to produce a bimetallic pipe comprising an inner CRA coat bonded to a structural steel [21–23]. Various strategies have been proposed for manufacturing these CRA-steel pipes, including expanded mechanical lined [24,25], hot-rolled metallurgically bonded clad plate [19], explosion-welded clad plate [26], and weld overlay cladding [27–33]. Among the aforementioned piping processing, the “weld overlay cladding”, which refers to welding processes where layers of

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metal are deposited onto a base metal to improve properties the base lacks, such as resistance to harsh, corrosive, and oxidising environments [21], has been successfully used in oil and gas for risers [34]. Smith [35] highlights that overlay welding methods capable of achieving the required chemistry in a single layer are crucial for overlaying high-nickel alloys, such as C276 and C22.

There are many fusion welding processes available for the manufacture of Ni-based cladding, mainly those with high levels of mechanisation and automation [36]. Submerged Arc Welding (SAW), Plasma Transferred Arc (PTA), and Gas Metal Arc Welding (GMAW) variants such as Cold Metal Transfer (CMT) and Gas Tungsten Arc Welding (GTAW), as well as Laser Cladding, are commonly used for cladding deposition. SAW offers high productivity, is cost-effective for large-scale applications, and supports the application of thick coatings [11,21,30–33,37,38]. However, high dilution can affect the coating, necessitating multiple layers; additionally, high heat input causes significant distortion and a wide HAZ, which is not ideal for low-thickness or thin coating layers [12,38,39]. The PTA process utilises a powder feedstock for the deposition of low-dilution layers, offering excellent wear/corrosion resistance, making it suitable for complex geometries [33]. Recent advances in wire feeding control have led to the development of the cold metal transfer (CMT) process, a variant of GMAW [32,37]. This enhanced short-circuit control results in reduced heat input, low dilution levels, minimal distortion and spatter, making it suitable for thin cladding and position welding, with the added benefit of cost-effectiveness [37]. A more advanced technology is laser cladding, which uses a high-energy concentrated laser beam to melt powders or wire feedstocks [38]. Its precise control and very low heat input result in clads with minimal HAZ extension and distortion, very low dilution levels, thus preserving the CRA properties.

GTAW with wire feeding already offers a high-quality weld due to precise control and the separation of arc power and wire feed, which can dramatically alter the melting rate [12]. This process is suitable for thin and moderate-thickness layers, as well as high-quality overlays. Its low cost-effectiveness is another appealing feature for the industry. Generally, it is considered a slow-deposition process, making it less productive for large areas compared with other methods. However, its high dilution in the first layer requires manufacturers to deposit two layers using both hot- and cold-wire variants to achieve the desired metallurgical properties. Therefore, optimising the welding procedure to produce a high-quality, defect-free clad with a minimum thickness of 3.0 mm in a single layer significantly affects cost and productivity by reducing labour time by at least half. Nonetheless, proper adjustment of operational conditions and welding parameters can increase the melting rate, leading to notable improvements in weld cladding [10–12].

Nonetheless, alloys with high Mo content often exhibit metallurgical problems under non-equilibrium conditions, such as microsegregation and precipitation—which occur in the fusion zone during weld pool solidification—thereby impairing the alloy's service performance [40–42]. Therefore, studies on these topics are essential, given the high susceptibility of Ni–Cr–Mo alloys to secondary-phase precipitation, which is strongly linked to solidification cracking [43] and corrosion susceptibility [44]. Manikandan et al. [45] investigated the suppression of microsegregation in Hastelloy C276 weldments using a pulsed GTAW current. According to the authors, joints produced with pulsed current exhibited superior mechanical properties, owing to a narrower weld with almost no heat-affected zone, a refined microstructure in the fusion zone, and reduced microsegregation. Haemers et al. [31] examined a Hastelloy C276 weld overlay produced by laser deposition on a steel substrate and confirmed efficient coating deposition. However, the author reported the formation of secondary phase precipitation in the weld metal. In another study, Pizzatto et al. [39] assessed NbC-reinforced Hastelloy C276 alloy composite coatings produced by laser cladding. The resulting microstructure included Ni- γ (FCC) dendrites with an interdendritic carbide network of lamellar-like M_6C and blocky-like conjugated MC - $M_{23}C_6$ carbide. The authors highlighted the

Table 1

Chemical composition of the base and filler metal.

Alloy	Chemical composition (wt.%)							
AWS ER NiCrMo-4 (Hastelloy C276®)	Ni	C	Cr	Mo	Fe	W	Mn	Si
	56.8	0.002	16.1	16.3	6.1	3.4	0.5	0.03
ASTM A516 Gr. 60	Ni	C	Cr	Mo	Fe	Al	Mn	Si
	0.01	0.15	0.02	0.01	Bal.	0.02	0.95	0.2

Table 2

Composition of shielding gases.

Shielding gas	Composition		
	Ar	He	NO
Ar + NO	99.98%	-	0.01%
Ar + He	70%	30%	-

significant influence of dilution on wear performance. Low heat input resulted in a higher carbide fraction and better wear resistance than high heat input, but it adversely affected wear performance due to high dilution.

Recent studies have evaluated Ni-based alloy weld metals, including Inconel 625 [30,33,46] and Inconel 686 [27–29,47]. Only limited research has been conducted on Hastelloy C276 welds [45,48–50]. Few studies have explored this alloy's dissimilar welds with steels [49,50]. However, even fewer address the weld cladding explicitly [50]. Therefore, given the potential of Hastelloy C276 for various industrial applications in harsh environments and its use as a cladding material, which remains insufficiently explored in the literature, this investigation is justified. Furthermore, manufacturing engineering challenges include assessing the cladding of Hastelloy C276 with optimised welding parameters to achieve a high-quality, single-layer coating. By examining the microstructural features and properties of dissimilar weld overlays of Hastelloy C276 on a C–Mn steel substrate, using optimised GTAW cold wire feed parameters to deposit a single-layer cladding, this research offers promising insights into the advantages of process optimisation, paving the way for more efficient and cost-effective welding procedures using GTAW cold wire feeding in the future.

2. Experimental procedure

The filler metal used in this study was AWS ER NiCrMo-4 (similar to Hastelloy C276), and the base metal was ASTM A516 Gr 60 steel; the chemical compositions of both metals are presented in Table 1. This steel was selected because it is used to construct boilers and pressure vessels for the petrochemical and oil and gas industries. Argon containing 30% helium and 0.01% NO was used as the shielding gas (Table 2) for GTAW (cold-wire feed) welding process.

The GTAW tests were conducted using an electronic power supply and a data acquisition system to monitor and record current and voltage during welding. The torch movements were performed on a robotic workbench to control welding speed and generate the weaving motion (torch oscillation) during the tests (Fig. 1a). An automatic GTAW wire feeder supplied the filler metal (Fig. 1b). A positioning unit that allowed for precise geometric adjustments guided the wire into the arc weld. Multiple pass welds were applied side by side to produce the weld cladding. All welds were performed flat on 200 mm × 150 mm × 12.5 mm steel plates.

The conditions evaluated were selected from an experimental matrix based on the Taguchi method, as previously reported [12]. These indicated that variables such as current, welding speed, heat input, and weaving (arc oscillation) significantly affected the geometry and dilution characteristics of the welds [12], necessitating a more detailed assessment of the impact of multipass welding itself.

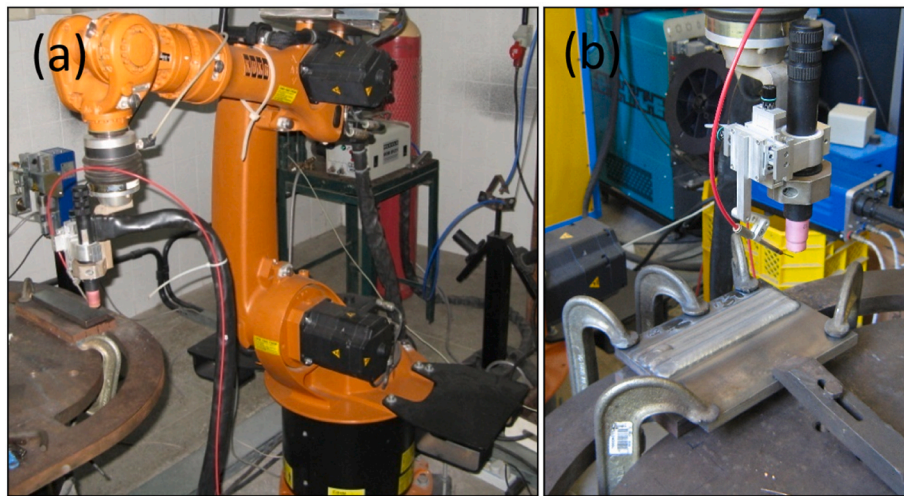


Fig. 1. (a) Robotic workbench with an integrated welding system. (b) The GTAW torch and wire feed positioner are shown in detail.

Table 3

Welding parameters are used to deposit the claddings.

Test	Shielding gas	Waving	RMS current I_{rms} (A)	RMS voltage U_{rms} (V)	Welding speed S_w (mm/min)	Heat input (kJ/mm)	Wire feed speed S_{wf} (m/min)
T1	Ar + NO	Triangular	334.2	20.9	250	1.1	6.0
T2	Ar + NO	Double 8	334.4	20.8	210	1.3	7.0
T3	Ar	Double 8	384.6	21.5	210	1.5	8.5

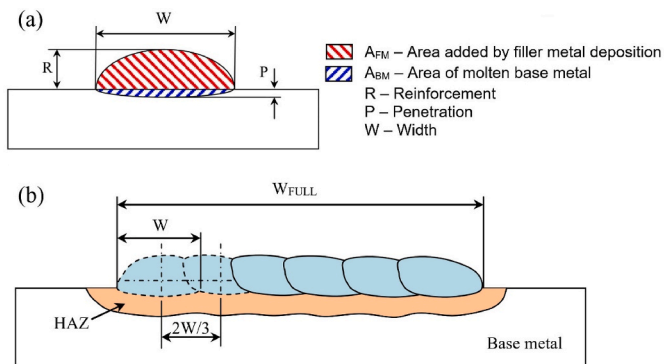


Fig. 2. (a) Representation of a single weld bead and the main geometrical features used to evaluate the deposition. (b) Drawing showing the weld bead overlap and consolidation of the cladding layer [12].

The welding heat input was determined using Eq. [1], where HI is the heat input, I is the RMS welding current, U is the RMS voltage, S is the welding speed and, η is the arc efficiency. As previously mentioned, the welding current level was fixed, and the different heat inputs were obtained by varying the welding speed. The main parameters selected for the welding discussed above are shown in Table 3. During welding, a calibrated data-acquisition system was used to measure and monitor current and voltage. A robotic workbench controlled the welding speed, while a calibrated system measured the wire-feed speed to ensure proper adjustment. The arc efficiency was assumed to be 0.65.

$$HI = \frac{IU}{S} \eta \quad \text{Eq. [1]}$$

The wire feed speed (S_{wf}) was proportional to the welding current and was not selected as an independent variable. Furthermore, the best conditions for reducing dilution were achieved when each arc power level reached its maximum feed rate. Other variables with a moderate effect, or whose prior results indicated optimal values for applying

overlays, were held constant in this study; for example, the electrode tip-to-workpiece distance (ETWD) was 10 mm.

Although this ETWD value is uncommon for conventional GTAW welding, a previous study [12] showed that the optimal value corresponds to the minimum melting of the base metal without compromising coalescence between the weld metal and the substrate. The distance from the tip of the wire to the workpiece (DTWW) was set to 3 mm, indicating a favourable geometry-dilution relationship. The weld beads were deposited side by side, overlapping one-third of the preceding bead's width, to produce the cladding layer (Fig. 2). Some minor parameters were maintained constant across all conditions, including the electrode tip angle (50°), the angle of attack of the wire feeder guide (50°), and the shielding gas flow rate (15 l/min). The wire feed speed was set to the maximum value for each condition, resulting in a low dilution level.

After welding, the dilution of the claddings was measured in two ways: one based on metallographic analysis of the weld metal cross-sectional area, and the other on chemical composition analysis. Samples of the test plates were cut for macro- and microanalyses. These samples were prepared according to standard metallographic procedures. Initially, they were ground with silicon carbide waterproof sandpaper, polished with a diamond paste to a final 1 μm grit, chemically etched in a 2% Nital solution for ASTM A516 Gr.60 steel, and finally electrolytically etched in a 10 wt% chromic acid solution at 2 V for 15–20 s. These samples were used for both dilution assessment and microstructural characterisation.

Image analysis software was used to determine the cross-sectional area of the weld overlay. In this cross-section, the area added by the deposition of the filler metal (nickel alloy) (A_{fm}) and the area corresponding to the base metal (steel) that melted during welding (A_{bm}) were measured. Thus, dilution was defined as the ratio of the melted steel area to the total weld overlay area ($A_{fm} + A_{bm}$), as given in Eq. [2]. The second method involved using energy-dispersive X-ray spectroscopy (EDS) to determine the chemical composition. Based on the results of the iron content in the weld metal, the dilution was calculated using Eq. [3], where D is the dilution and Fe_{wm} , Fe_{fm} , and Fe_{bm} are the iron contents in

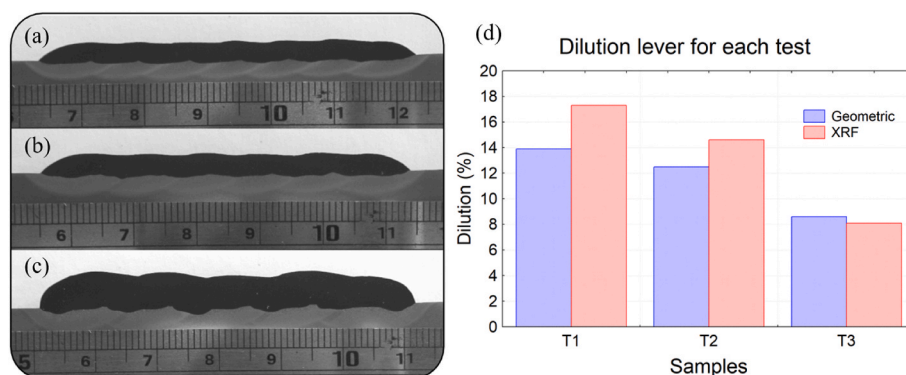


Fig. 3. Cladding cross-sections of macrostructures deposited under different conditions: (a) Sample T1 (1.1 kJ/mm); (b) Sample T2 (1.3 kJ/mm); and (c) Sample T3 (1.5 kJ/mm). (d) Bar plot of the dilution levels.

%wt of the weld metal, the filler metal, and the base metal, respectively.

$$D = \frac{A_{bm}}{(A_{fm} + A_{bm})} \cdot 100 \quad \text{Eq. [2]}$$

$$D = \frac{Fe_{wm} - Fe_{fm}}{Fe_{bm} + Fe_{fm}} \cdot 100 \quad \text{Eq. [3]}$$

Two scanning electron microscopes (SEMs) were used to examine the microstructures: a Phillips XL30 model and a Carl Zeiss EVO 40. The Everhart-Thornley detector was employed to produce secondary electron (SE) images. Both SEMs were equipped with energy-dispersive X-ray spectroscopy (EDS) systems and were operated at 20 kV. Samples for characterisation were examined by transmission electron microscopy (TEM) and scanning transmission electron microscopy (STEM) in both bright-field and dark-field modes. The TEM/STEM analyses were performed using a JEOL JEM 2100 ARP microscope coupled to an EDS microanalysis system. The samples were extracted and prepared in the plain view section.

X-ray diffraction (XRD) characterised the secondary phases precipitated during solidification and solid-state transformation. This evaluation followed ASTM E963, which covers procedures for isolating TCP (topologically close-packed) phases in nickel-based alloys. A Panalytical XPert Pro MPD X-ray spectrometer was used in the present investigation. X'Pert HighScore software and its database were used for postprocessing and peak matching. The scan details used in these analyses were as follows: the 2θ range was 30–90°, with a step size of approximately 0.02°, and counting was performed for 4 s at each step. Cu K α radiation was used, with the X-ray generator set to 40 kV and 40 mA.

Microhardness tests across the weld interface, from HAZ to FZ, were performed in accordance with ASTM E384-99 [51], using a 100 gf load applied for 15 s, and 0.2 mm spacing. Immersion corrosion testing, according to ASTM G48 [52], method C, was performed in duplicate to determine the critical pitting temperature (CPT), as outlined in various international standards. A ferric chloride solution (6% FeCl₃ in 1% HCl) was used. Samples measuring 25 × 50 mm and 3.0 mm thick were ground with waterproof silicon carbide sandpaper up to 120 mesh and immersed in a beaker containing 600 ml of acidified chloride solution. The samples were supported by a glass stand that contacted only 4 points on the sample edges. Before and after each test, the surfaces were visually inspected, and the mass was measured to detect pitting or mass loss exceeding 0.0001 g/cm². The beaker containing the solution was placed on a heating plate, and an initial temperature of 65 °C was set, increasing by 5 °C every cycle up to 85 °C.

Table 4

The dilution level for each test condition was determined by different techniques.

Dilution (%)		
Sample	Geometric	XFS
T1	13.9	17.3
T2	12.5	14.6
T3	8.6	9.7

3. Results

3.1. Dilution and chemical composition

Fig. 3 displays the macrostructures of the cladding cross-sections. The images show no cracks or macroscopic defects, such as pores or gas bubbles, under any of the tested conditions. Additionally, Fig. 3 indicates that the different welding conditions employed in the study resulted in varying levels of dilution.

The dilution level for each test condition (Table 4) was determined from the chemical composition of the weld claddings (Table 5), as measured by X-ray fluorescence spectroscopy (XFS) and geometric cross-section analysis. According to Table 4, Condition T1, with the lowest heat input, showed the highest dilution. Conversely, T3, with the highest heat input, exhibited the lowest dilution level. However, it is essential to note that these welds were also performed with increased heat input by reducing the welding speed and increasing the wire feed rate. These adjustments significantly increased the weld pool volume. Therefore, the dilution behaviour observed in this study is closely related to the combination of heat input and molten metal volume during welding. To explain this behaviour, the barrier effect caused by the molten metal between the electric arc and the substrate must be considered. Reducing welding speed led to an accumulation of liquid metal, thereby increasing the weld pool size. A larger weld pool acts as a barrier to the arc on the substrate, resulting in less heat being transferred directly to the steel and, consequently, less substrate fusion. The chemical composition of the weld claddings (Table 5) was measured using X-ray fluorescence spectroscopy (XFS). This table lists the primary chemical elements of the alloys in question, along with any secondary elements present in small quantities.

The iron assessment results show that iron ranged from 15.1 wt% to 22.0 wt%, indicating significant variation in dilution, as previously observed. It is important to note that the pure alloy C276 already contains approximately 6.0 wt% iron, which explains the high iron content in the weld overlays. The chromium content ranged from 13.4 to 16.3 wt%, while the molybdenum content varied between 13.7 wt% and 15.2 wt%. Tungsten (W) content was 2.7 wt% for Condition 1, which had the highest dilution; however, it increased to approximately 3.7 wt% as the

Table 5
Chemical composition of the weld claddings.

Chemical composition wt.%										
Sample	Ni	Cr	Mo	Fe	W	Mn	Si	Co	V	C*
T1	45.0	13.4	13.7	22.0	2.7	0.67	0.36	0.11	0.12	0.035
T2	44.9	15.5	14.2	19.6	3.1	0.72	0.19	0.12	0.22	0.031
T3	44.4	16.3	15.2	15.1	3.7	0.90	0.38	0.10	0.10	0.025

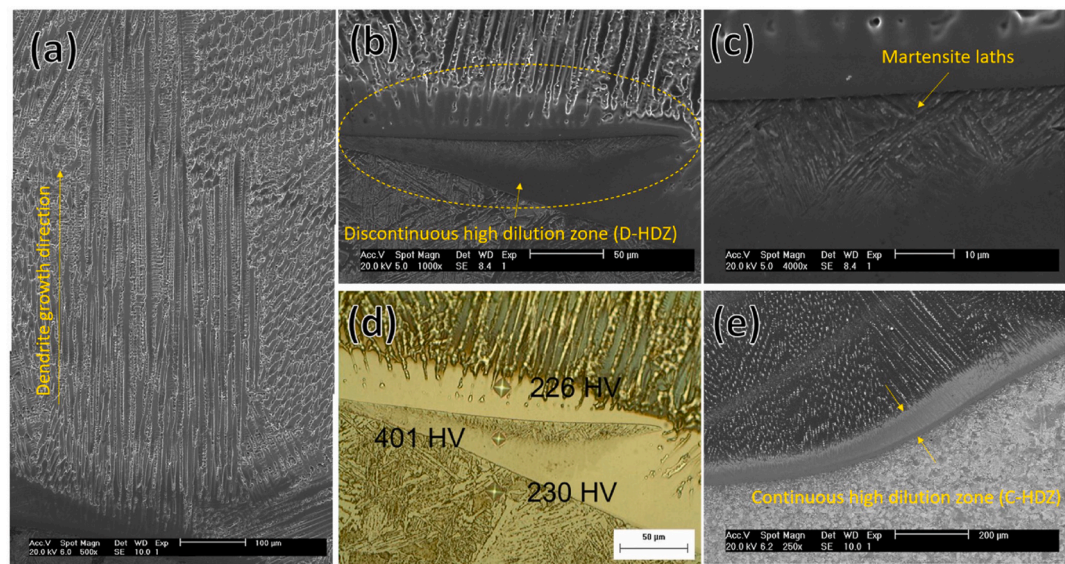


Fig. 4. Typical microstructure along the fusion boundary of the T3 sample. (a) Microstructure of Hastelloy C276 showing evolution of the S/L interface. (b) Example of a D-HDZ. (c) Enlarged image showing martensite laths within the D-HDZ. (d) Same as Fig. 4b after the microhardness test, showing increased hardness. (e) Example of a C-HDZ.

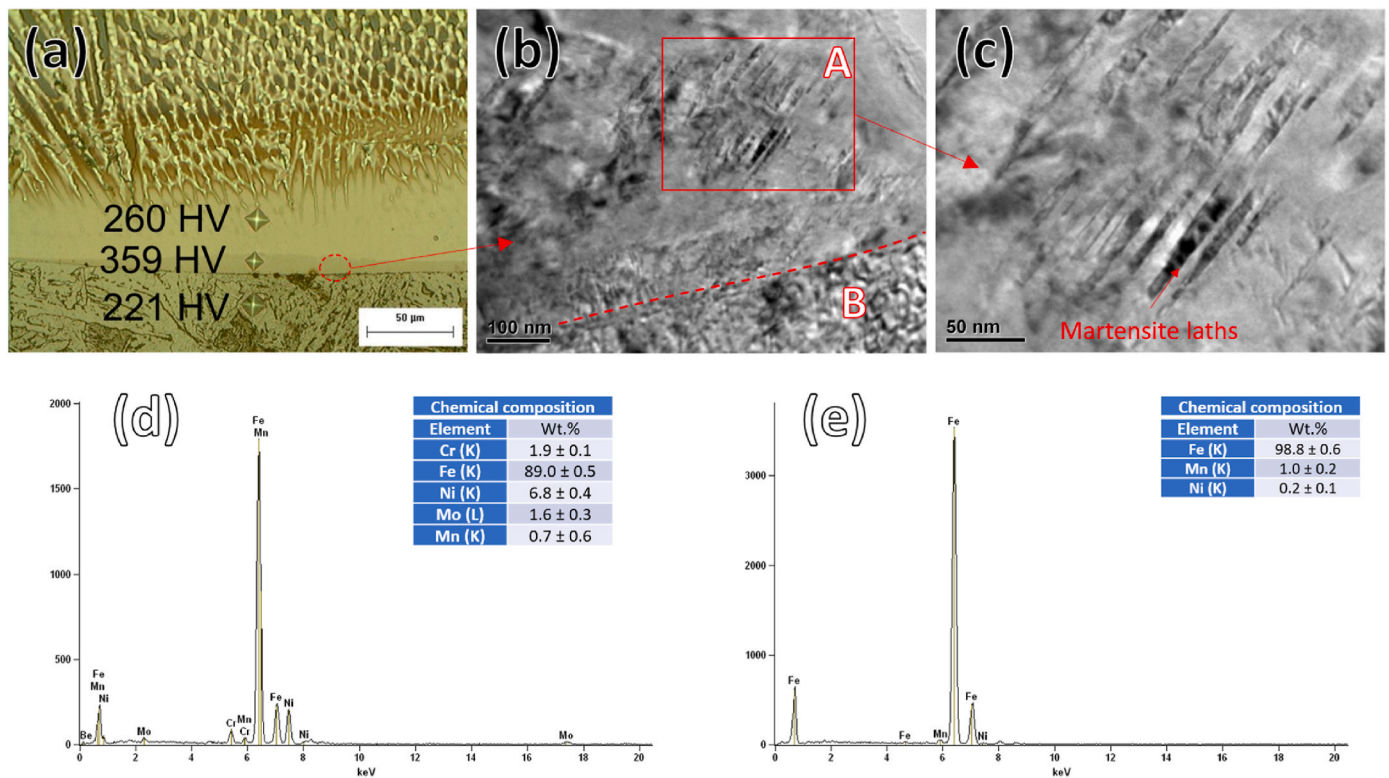


Fig. 5. (a) Discontinuous fine grey region along HDZ with high hardness. (b) TEM BF image along the interface showing nanometric martensite laths. (c) Enlarged view from the red rectangle. (d) EDS analysis and chemical composition of upper region “A”. (e) EDS analysis and composition of base metal in area “B”.

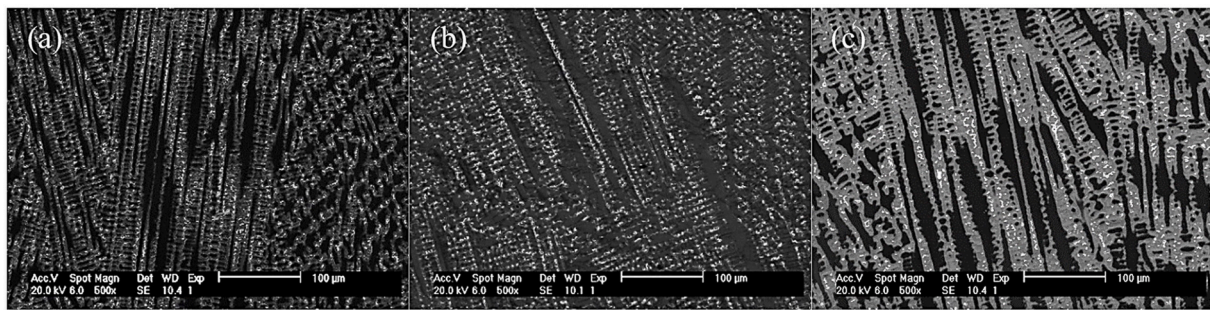


Fig. 6. The as-welded microstructure of the Hastelloy C276 weld metal for the three heat inputs was evaluated. (a) T1 sample welded at 10 kJ/cm; (b) T2 sample welded at 12 kJ/cm; (c) T3 sample welded at 15 kJ/cm.

dilution decreased. Some secondary elements, such as manganese, aluminium, and cobalt, were also present at very low concentrations, and the carbon content was estimated by dilution.

3.2. Microstructural characterisation

This section presents the main results obtained using various microscopy and microanalysis techniques, aimed at understanding the multiple changes at the coating-substrate interface and in the fusion zone of the Hastelloy C276 alloy diluted with carbon steel. Fig. 4a shows the evolution of the solidification front from the fusion line. A faint region of planar growth can be observed, resulting from an intense thermal gradient (G) associated with a low solid growth rate (R) (high G/R). This soon evolves into a cellular solidification front and then transitions to dendritic columnar growth as the G/R ratio gradually decreases. At the interface, a pronounced chemical variation is observed, termed the High Dilution Zone (HDZ), as shown in Fig. 4b and e. Some of these zones exhibit significant chemical heterogeneity, distributed discontinuously (D-HDZ), as they are regions where convection fronts meet. These typically result in zones with high substrate iron concentration and some alloying-element insertion from the filler metal, thereby conferring high hardenability. This is shown in the inset of Fig. 4c, where an essentially martensitic microstructure is observed. The measured hardness values for these regions are significantly higher than those of the fused zone and the adjacent HAZ, as seen in Fig. 4d. Although critical because they

act as a localised brittle zone (LBZ), these regions correspond to a small fraction of the total interface, with the majority consisting of a continuous HDZ (C-HDZ), with a uniform chemical gradient ranging from the base metal composition to the fused zone composition, as shown in Fig. 4e.

In some samples, greyish regions were also observed along the interface. These regions were not revealed by the applied chemical and electrochemical etchings, remaining apparently single-phase (Fig. 5a). However, transmission electron microscopy (TEM) analyses of the interface region revealed nanometric martensitic laths, as shown in Fig. 5b–c. The EDS in the areas noted as “A” and “B” in Fig. 5b are presented in Fig. 5d–e. The results indicated that the composition above the dotted red line, corresponding to the fusion line, contained alloying elements such as Ni, Cr, and Mo. The presence of these elements evidences migration from the molten zone, possibly by diffusion into the liquid. The entry of these elements into the Fe-rich liquid base metal layer, which is stagnant near the fusion line, contributes to the stabilisation of austenite at temperatures below the liquidus temperature and significantly increases hardenability. This favours the formation of martensite laths when such regions reach temperatures below M_s transformation temperature. Region B, however, retains the typical carbon-steel composition, showing no trace of alloying elements, confirming that it is the base metal beneath the fusion line.

Continuing the analysis toward the Hastelloy C276 fusion zone, it displayed a columnar-dendritic pattern with a nickel-fcc matrix and

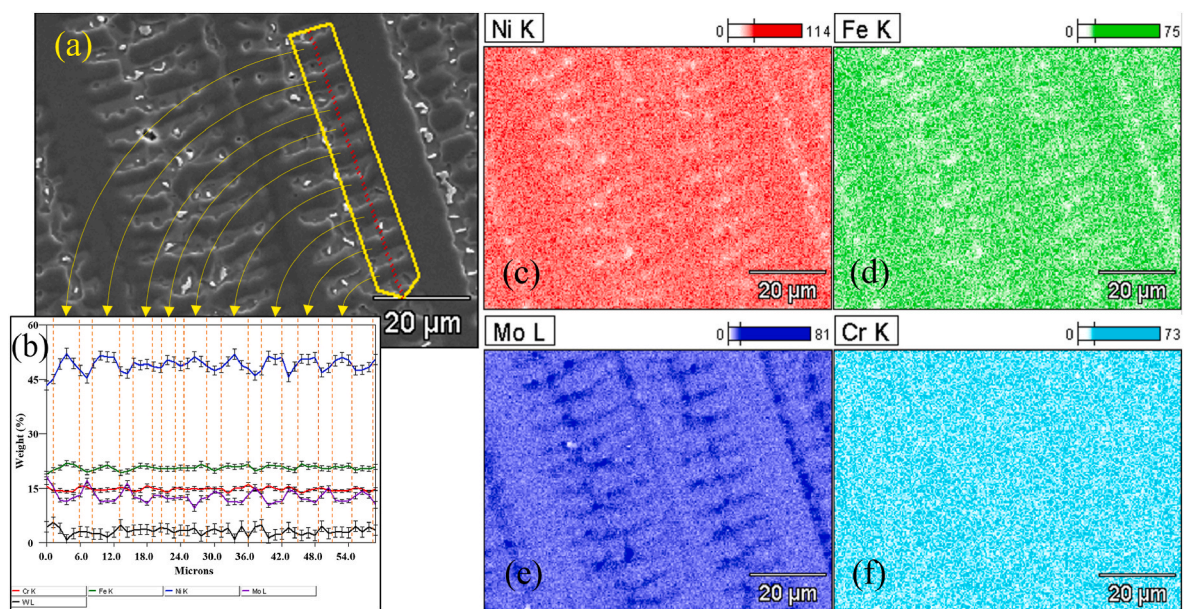


Fig. 7. (a) Microstructure of the T1 sample showing a dendritic solidification pattern in the weld metal and an indication of EDS line scan analysis. (b) Chemical composition profile along the secondary dendritic arms obtained by EDS. Elemental mapping of the (c) Ni, (d) Fe, (e) Mo, and (f) Cr alloying elements.

Table 6

The concentrations of the alloying elements in the dendritic core and interdendritic region were measured via EDS and the partitioning coefficient (k).

Elements	Sample T1			Sample T3		
	C_s	C_l	K	C_s	C_l	K
Ni	47.8	45.1	1.06	50.6	48.6	1.04
Cr	14.1	13.9	1.01	16.3	15.8	1.00
Mo	10.9	14.5	0.79	13.1	17.9	0.86
W	2.7	2.9	1.00	3.6	3.4	0.97
Fe	24.6	23.4	1.10	16.0	14.5	1.07

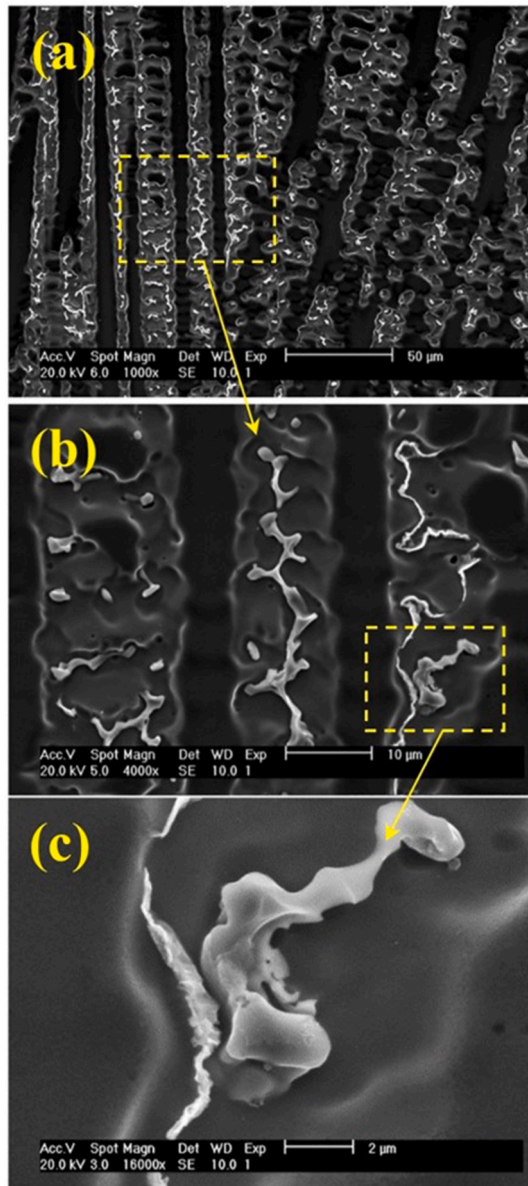


Fig. 8. (a) Typical as-welded microstructure of the T2 weld cladding. (b) Enlarged image highlighting the dendritic pattern and the secondary phases formed along the interdendritic. (c) In detail, an example of a shapeless secondary phase.

secondary phases in the interdendritic regions, as shown in Fig. 6. Film-like precipitates were observed along the grain boundaries. Element distribution reveals microsegregation: nickel and iron concentrate in the dendrite core, while molybdenum is highly segregated to the interdendritic region, as shown in the EDS map in Fig. 7. No significant changes

were observed in the distributions of chromium and tungsten.

Fig. 7a shows the SEM image with EDS lines intersecting the secondary dendrite arms from the T1 sample, highlighted by arrows pointing to the dendrite cores. Fig. 7b shows the alloying element profile, revealing that interdendritic regions are depleted in nickel and iron, while the dendrite cores are enriched. Molybdenum concentration increases in the interdendritic regions but depletes in the dendrite core. Variations in profiles for other elements, like chromium and tungsten, are minimal. EDS maps for Ni, Fe, Cr and Mo are presented in Fig. 7c–f to graphically highlight the preferred elemental partitioning. EDS measurements of the last weld bead analysed the alloying element concentration between the dendrite core and interdendritic region. From these measurements, the distribution coefficient (k) for key alloying elements in the Hastelloy C276 was determined, as shown in Table 6. Nickel and iron had k values greater than 1 ($k > 1$), indicating that they were quickly incorporated into the solid during solidification, thereby depleting the liquid between the solid dendrites. Molybdenum showed $k < 1$, suggesting it segregates toward the liquid, while chromium showed k near 1. Tungsten also appeared to be biased toward the liquid, but the wide dispersion in percentages and a k close to 1 make it difficult to confirm this definitively.

Microsegregation leads to secondary phase formation during the final stage of solidification. Fig. 8 highlights the microstructural features of the columnar-dendritic growth zone, where SEM micrographs reveal secondary phases in the interdendritic regions, taken from the T2 sample. These shapeless particles arise from microsegregation of alloying elements, notably molybdenum, which alters the liquid's composition and thermodynamic equilibrium, thereby inducing a new phase. During the microstructural analysis, various morphologies, including shapeless forms, were observed. Film-like precipitates formed along solidification grain boundaries, particularly near the clad/substrate interface, and were primarily observed on reheated weld beads due to multiple passes.

Fig. 9 shows a topographic view of film-like precipitates characterised by roughness and protrusions along their boundaries, likely resulting from an eutectic reaction during solidification. These finger-like shapes may indicate interlacing with the matrix or precipitate growth from reheating. Lamellar morphology particles, with thin plates aligned in a preferred direction with the γ -fcc matrix, were also noted (Fig. 7c). Silva et al. [27] reported a similar morphology in their study of alloy 686 used as cladding on carbon steel.

Fig. 10 shows segmented images obtained from the SVRNA software [53] for various welding conditions. Green indicates the dendrite core, yellow the interdendritic region, and blue marks the secondary phases. The images illustrate that nickel dendrite growth increases with rising heat input during solidification. A statistical analysis for the final weld pass showed an increase in secondary phases with higher heat input (Fig. 11, Table 7), and visual inspection noted larger particle sizes at elevated heat inputs.

Fig. 12 shows EDS chemical mapping of a precipitate taken from the T3 sample, revealing decreased nickel and iron contents, whereas molybdenum content increased significantly. Chromium distribution remained consistent, with no preference for the matrix or the precipitate. Tungsten, present at low levels, showed no preferential distribution in EDS mapping, although spot analyses indicated an increased concentration in the precipitate. These findings suggest that secondary phases formed during the solidification of the weld overlay on the alloy C276 are predominantly molybdenum-rich.

TEM analyses were used to characterise the precipitated secondary phases. Fig. 13 shows a bright-field TEM image of an elongated secondary phase, resembling the precipitates observed earlier by SEM in interdendritic regions. Imperfections, akin to grain boundaries, suggest that these elongated phases consist of multiple crystals (Fig. 13b). The selected-area diffraction (SAD) pattern of the γ -fcc matrix is shown in Fig. 13c. Fig. 13d reveals that the precipitate is a P -phase with an orthorhombic crystal structure (Pbnm space group) and lattice parameters $a = 0.907$ nm, $b = 1.701$ nm, and $c = 0.474$ nm. EDS analysis of the

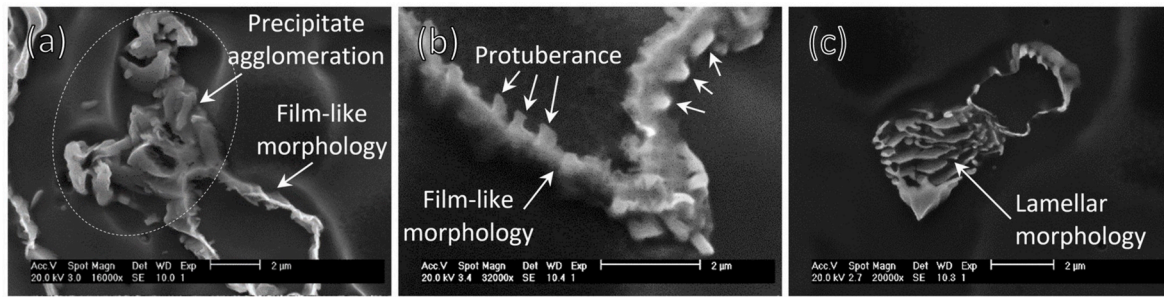


Fig. 9. Topographic aspects of some secondary-phase precipitates. (a) Film-like and agglomerate precipitate morphologies. (b) Film-like with protuberances. (c) Lamellar precipitate morphology.

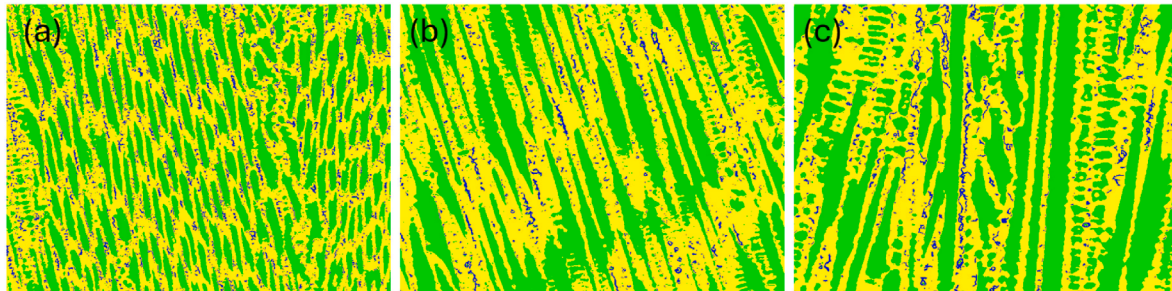


Fig. 10. Segmented image obtained by image analysis with SVRNA software: (a) Sample T1; (b) Sample T2; (c) Sample T3.

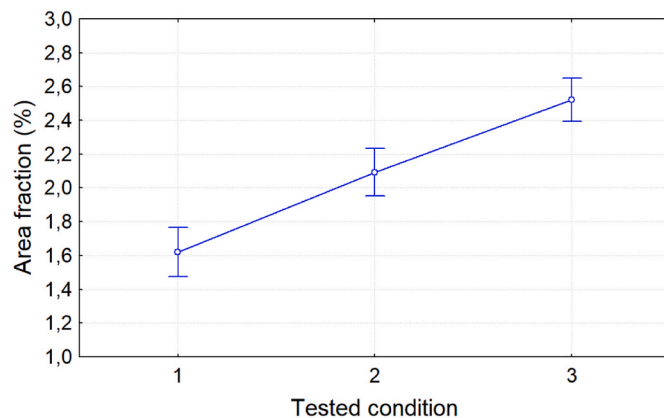


Fig. 11. Statistical results for the area fraction of secondary-phase precipitation.

Table 7
Results of the quantitative analysis of the secondary phases precipitated in the weld metal.

Sample	Precipitate content (%)
	Last pass
T1	1.62 ± 0.3
T2	2.09 ± 0.4
T3	2.52 ± 0.3

matrix and the precipitate is shown in Table 8 and indicates that *P*-phase contains 20.4 wt% nickel, along with chromium (10.9 wt%), iron (10.2 wt%), molybdenum (51.1 wt%) and tungsten (7.5 wt%). TEM analysis was conducted on samples extracted from the last weld bead because the smaller precipitates were unaffected by reheating.

X-ray diffraction analyses of the extracted precipitates are presented in Fig. 14 and revealed a pronounced predominance of the *P* phase,

indicating Mo segregation during solidification. This phase is the primary secondary phase precipitated in the weld overlays microstructure. Peaks for the μ - and σ -phases were also identified, though some peaks overlapped between the *P*-phase and other phases.

3.3. Properties evaluation

In terms of properties, the analyses consisted of microhardness profiles across the interface, taken from HAZ to the FZ. The results are presented in Fig. 16, considering all tested conditions. In each graphic, two profiles were overlapped, one representing the first weld bead, and the other representing the last weld bead. In general, the average hardness in the HAZ decreased with increasing heat input. Peaks of high hardness were typically observed near the interface, resulting from HDZ formation. The profile, along with the fusion zone, indicated a hardness of 200–300 H V. This result reinforces the importance of the fine interface region (HDZ), which is approximately 0.5 mm in width.

The critical pitting temperature (CPT) results for samples consisting solely of the coating layer, tested under two dilution levels (low and medium), are shown below. It should be noted that the samples were sanded, washed, and weighed immediately before immersion; no pickling was carried out, and the tests were conducted in duplicate. It was found that samples immersed in ferric chloride and hydrochloric acid for 72 h at 65 °C exhibited no pitting or mass loss (Fig. 17). For samples immersed at 70 °C, macroscopic pitting occurred, and significant mass loss was observed for those with low dilution levels. The samples at medium dilution showed no pitting on visual inspection or mass loss. At 75 °C, both tested conditions resulted in pitting. Therefore, the CPT for the low-dilution condition was 70 °C, whereas for the medium-dilution sample it was 75 °C. To better understand the material's corrosion resistance, a final test was conducted at the maximum temperature specified in the standard (85 °C). The results indicated that both conditions failed, resulting in significant sample degradation. However, in this test, the mass losses experienced by the medium dilution samples were substantially greater than those observed in the low dilution samples, as seen in Fig. 17.

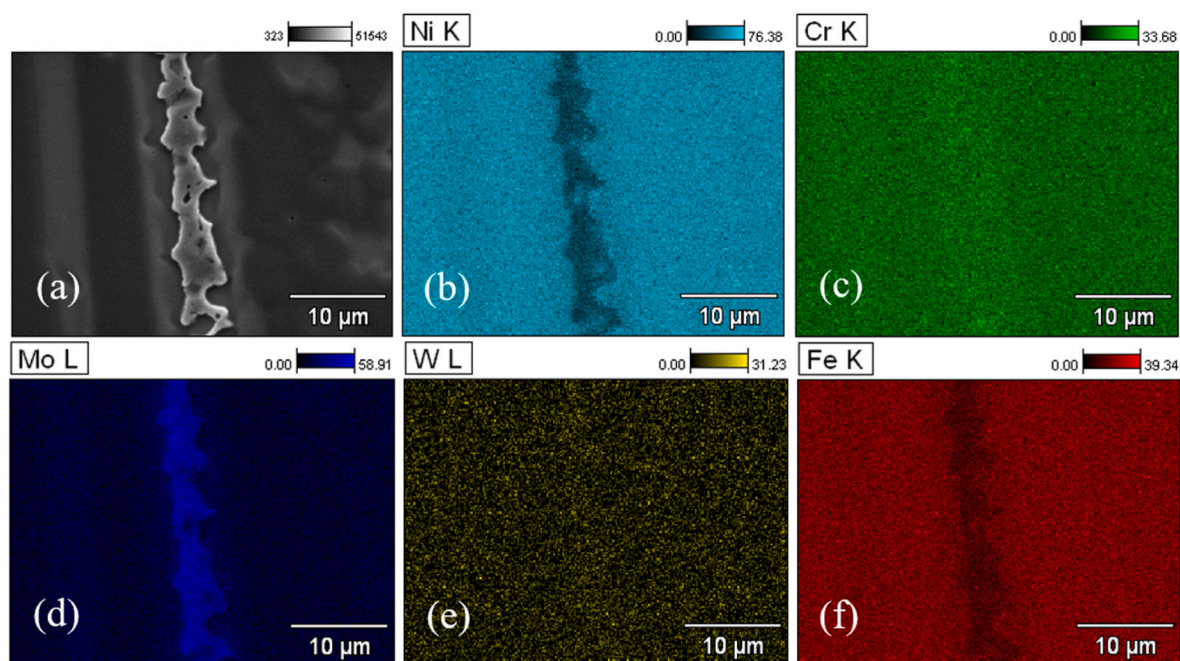


Fig. 12. Chemical mapping of a Mo-rich secondary phase formed in the interdendritic region of the T3 sample. (a) SEM image in SE mode; (b) Ni K; (c) Cr K; (d) Mo L; (e) W L; (f) Fe K.

4. Discussion

In cold-wire GTAW welding overlay, increased heat input, together with the wire melting rate, significantly reduced dilution, as indicated by XRF measurements, which decreased from 17.3% to approximately 9.7%. Geometric measurements reported dilution percentages of 13.9%, 12.5%, and 8.6% at higher heat input. Achieving less than 5.0% dilution in single-layer deposition was not feasible. Silva et al. [12] utilised the Taguchi method to optimise parameters to minimise dilution in Ni-based alloy overlays, achieving 5.5% dilution with carbon steel.

The iron content in the deposited layer may exceed 5.0 wt% because the filler metal contains around 6.0 wt% iron, despite low dilution. This is significant because international standards for welded coatings limit surface Fe content to 5.0 wt% to ensure resistance to pitting corrosion [54], particularly for the alloy AWS ER NiCrMo-3 (Inconel 625). Different alloys have varying concentrations of alloying elements (Cr, Mo, W) that impact the stability of the passivating layer. Thus, slightly higher Fe levels might be acceptable to maintain corrosion resistance. The limited literature on Fe content in solid-solution-strengthened Ni-based alloys underscores the need for further research in this area. Although this study found that increasing heat input and wire feeding speed decrease dilution, higher heat input also slows solidification, allowing greater segregation of alloying elements and promoting a faster solid growth rate, as evidenced by thicker dendrites in the micrographs.

In comparing lower and higher heat-input conditions for microsegregation, the partition coefficient decreased with increasing heat input. This may seem counterintuitive, as higher heat input typically results in a slower solidification rate and greater solute segregation. However, in dissimilar welding, the increase in Fe concentration from dilution plays a significant role. At lower heat input, higher Fe content reduces the amount of alloying elements available for segregation. On the other hand, it tends to reduce the solubility of Ni-alloying elements, thereby enhancing nickel's ability to reject alloying elements such as Mo. In contrast, higher heat input reduces dilution and Fe content, thereby increasing the concentration of solute available for segregation; however, Ni solubility is enhanced, thereby balancing the potential for increased segregation. Miná et al. [29] observed similar trends in the behaviour of Ni–Cr–Mo–W alloy Inconel 686, noting that Fe and Mo

segregation increased with dilution. At the same time, Cr and W showed slight changes in their partition coefficients.

Some authors have reported that microsegregation negatively impacts the mechanical response of welds. Manikandan et al. [45,55] observed that the strength and ductility of alloy C276 welds are compromised by Mo microsegregation. They also found that the fusion zone exhibited lower toughness than the base metal, which they attributed to the same issue. Solute segregation resulted in a slight increase of approximately 1% in secondary phases, corresponding to a 67% increase from the lowest to the highest heat input. Our primary question is whether this increase affects the overlay's corrosion resistance. These secondary phases, formed by elongated chains of precipitates in the interdendritic regions, are the one significant contribution of our investigation. Understanding their formation and impact on coating properties is vital to our research.

Understanding the evolution of secondary phases during weld solidification in complex alloy compositions is crucial for determining properties and behaviour. The Scheil–Gulliver model effectively outlines phase transformation as a function of temperature and solid fraction. The Scheil diagram (Fig. 15) generated with Thermo-Calc (Ni database version 8) indicates that the Ni-FCC matrix forms first and evolves until approximately 97% solidification, with Mo rejected by the remaining liquid. As the liquid fraction decreases, changes in composition and thermodynamics favour the nucleation of new phases. Elements like Cr and W remain concentrated in the liquid, leading to the potential formation of phases rich in Mo, Cr, and W, along with predictions of M_6C carbides. However, these were not identified in this study.

As shown by chemical analysis, many alloying elements, such as iron and tungsten, were present in the P phase, which has the stoichiometric formula $Cr_9Ni_{20}Mo_{21}$. Other researchers have also observed similar behaviours in commercial alloys containing tungsten and iron [27,40,41, 56,57]. Cieslak et al. [40] studied the solidification of this alloy. They observed that solidification begins with the crystallisation of the Ni-fcc phase, and that the P-phase forms at approximately 1285 °C. Perricone et al. [56] also evaluated the solidification path of Hastelloy class alloys and concluded that the solidification process of the alloy occurs following the same path previously described.

Regarding the formation of other secondary phases, such as the

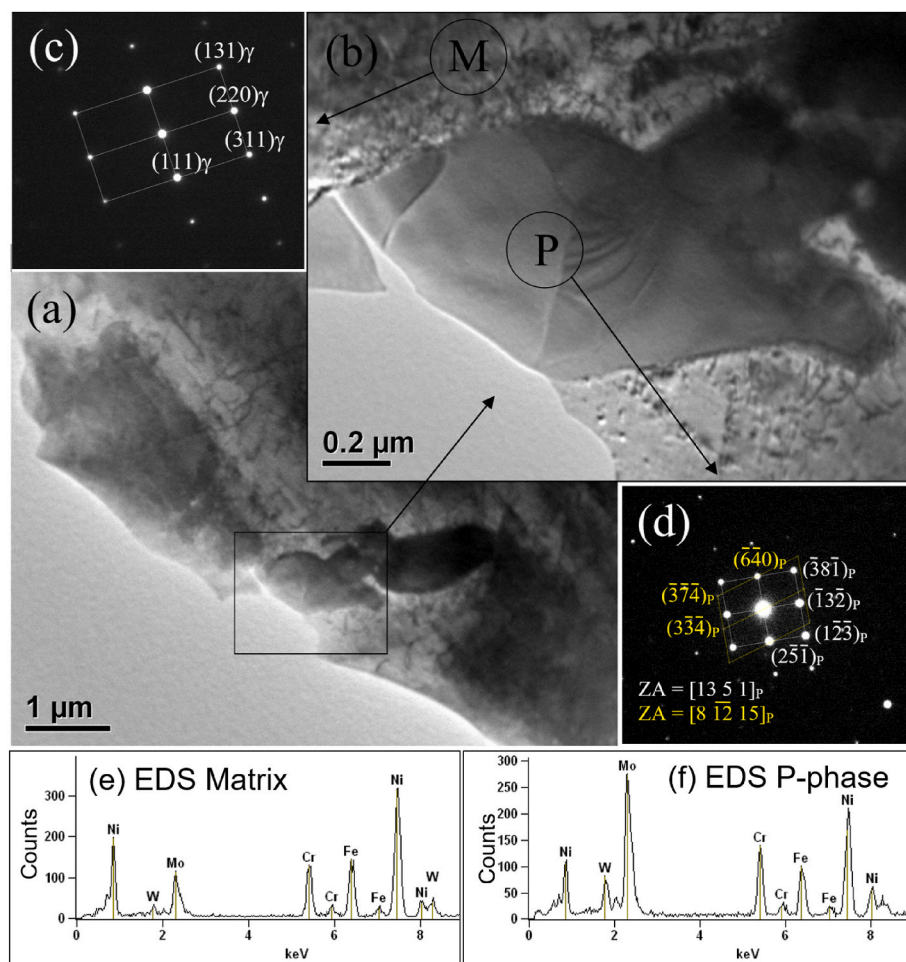


Fig. 13. (a) TEM bright-field image of a secondary phase precipitated in the interdendritic region of Sample T3. (b) Enlarged image of the secondary phase identified as the *P*-phase ($\text{Cr}_9\text{Ni}_{20}\text{Mo}_{21}$); (c) SAD pattern of the γ -FCC matrix under the axis zone $[211]$; (d) SAD pattern of the *P*-phase with two overlapping patterns under the axis zone $[13\ 5\ 1]_p$ and $[8\ 12\ 15]_p$; (e) EDS spectrum of the γ -FCC matrix; (f) EDS spectrum of the *P*-phase.

σ -phase, Cieslak et al. [40] and Perricone et al. [43,56] highlighted that the σ -phase is unlikely to form during the solidification of alloy C276 due to the low chromium content. However, the alloy conditions evaluated in the present study differed from those in the earlier studies [40, 41,43]. In our research, the filler metal was diluted with low-carbon steel as the base metal. The dilution will result in weld metal with a chemical composition different from that of the original C276 filler metal, significantly increasing the iron content. Therefore, σ -phase formation can result from iron incorporation due to dilution.

μ -phase has been related to a solid-state transformation caused by several welding thermal cycles, as reported by Raghavan et al. [41] and Cieslak et al. [40], transforming from the *P*-phase. Silva and colleagues [27] utilised transmission electron microscopy (TEM) to identify the coexistence of μ - and *P*-phases. The phases were observed to form together and exhibit a crystalline orientation relationship, suggesting similar behaviour. However, it can also result from an inhomogeneous distribution of the alloying elements into the confined liquid metal between dendrites. Cheng et al. [58] also found μ - and *P*-phases coexisting in Hastelloy C276/304 L dissimilar joints subjected to laser welding.

Additionally, the chemical composition varies, increasing from 6.0 wt% iron in pure alloy C276 to 15.1–22.0 wt%, depending on condition. A high iron content in the weld metal will reduce the overall concentration of molybdenum and tungsten, which are crucial phase-forming elements of the *P*- and μ -phases. Furthermore, increasing iron content in the Ni-fcc matrix can reduce the solubility of elements such as molybdenum, tungsten, and chromium. Therefore, due to stronger

segregation during solidification, the concentrations of iron, chromium, and molybdenum in some interdendritic regions are likely to promote the σ -phase, given that the weld metal is heterogeneous on a microscopic scale. This behaviour explains the presence of some peaks of the σ -phase observed in the diffractograms.

Regarding properties, coatings generally demonstrate good ductility. Since they are not included in structural calculations, their strength may be lower than that of structural steel; therefore, these properties were not assessed. Regarding hardness, a significant change occurred at the steel-C276 alloy interface, attributable to the complex microchemical and microstructural characteristics of this zone. This led to the formation of a high-dilution zone (HDZ) [59], in which the base metal, particularly Fe, significantly influences the local composition. Consequently, regions predominantly composed of Fe may stabilise austenite, depending on the concentration of alloying elements introduced by the Ni alloy. However, in situations where the concentration of Ni and other alloying elements is low and insufficient to stabilise the fcc range, the material will locally exhibit high hardenability, transforming into martensite during cooling and increasing the hardness of the interface. This is particularly critical in discontinuous zones, as shown in Fig. 16. Furthermore, the formation of these zones is affected by processing conditions, such as welding energy, which influences pool convection, the volume of metal added, the solidification rate, and the thickness of the stagnant liquid layer (boundary layer), among others. Analysing interface behaviour remains a topic requiring further exploration.

In terms of pitting corrosion resistance, the CPT temperature

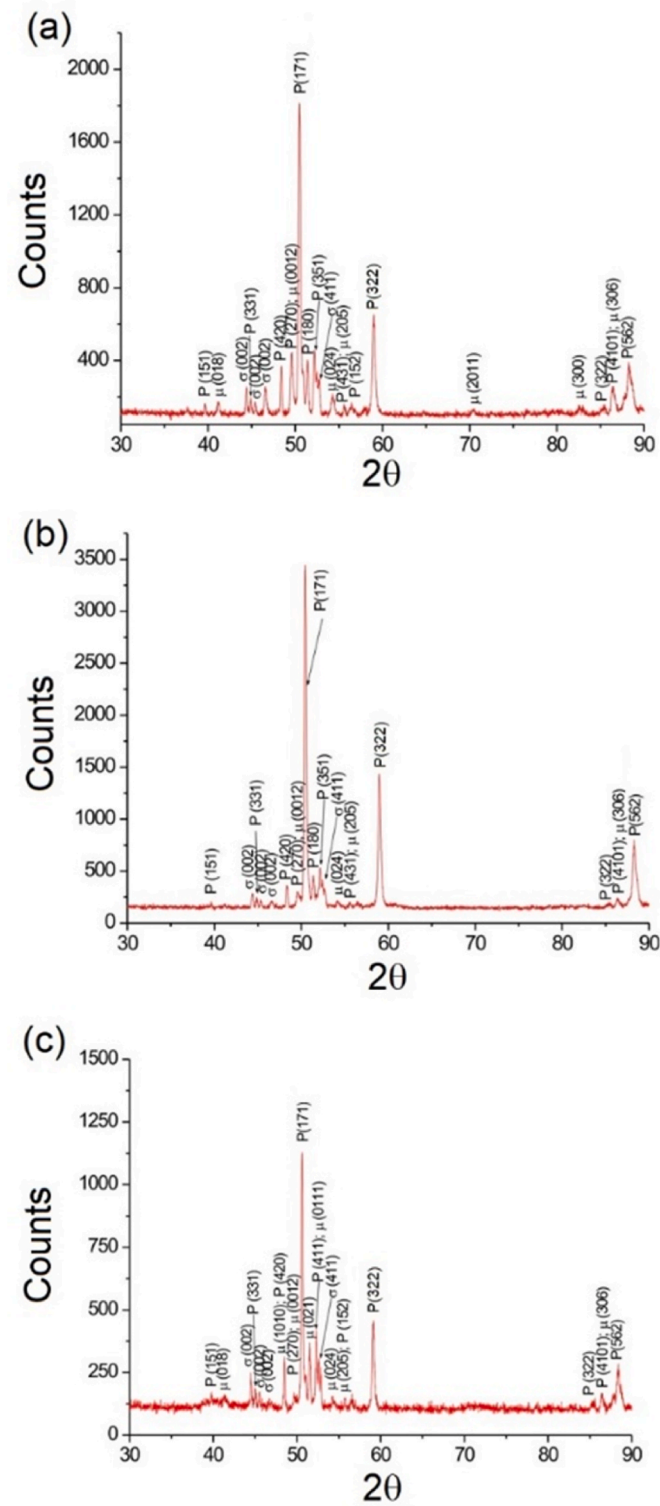


Fig. 14. X-ray diffraction spectrum of powders extracted according to the precipitation extraction technique. (a) Sample T1; (b) Sample T2; and (c) Sample T3.

achieved by the weld overlay with low dilution was lower than that with medium dilution. However, as shown in the mass-loss evolution graphs as a function of test temperature, although pitting occurred at a lower temperature, the mass-loss progression under this condition was slower. Compared with the medium-dilution condition, the latter took slightly longer to initiate pit formation; however, once nucleated, the corrosion

Table 8
Chemical composition of the EDS spectra of the matrix and the P-phase.

	Content (wt.%)				
	Ni	Cr	Mo	W	Fe
Ni matrix	40.3 ± 1.0	13.5 ± 0.4	23.1 ± 1.2	5.7 ± 1.0	17.4 ± 0.7
P-phase	20.4 ± 0.7	10.8 ± 0.3	51.1 ± 1.5	7.5 ± 0.9	10.2 ± 0.3

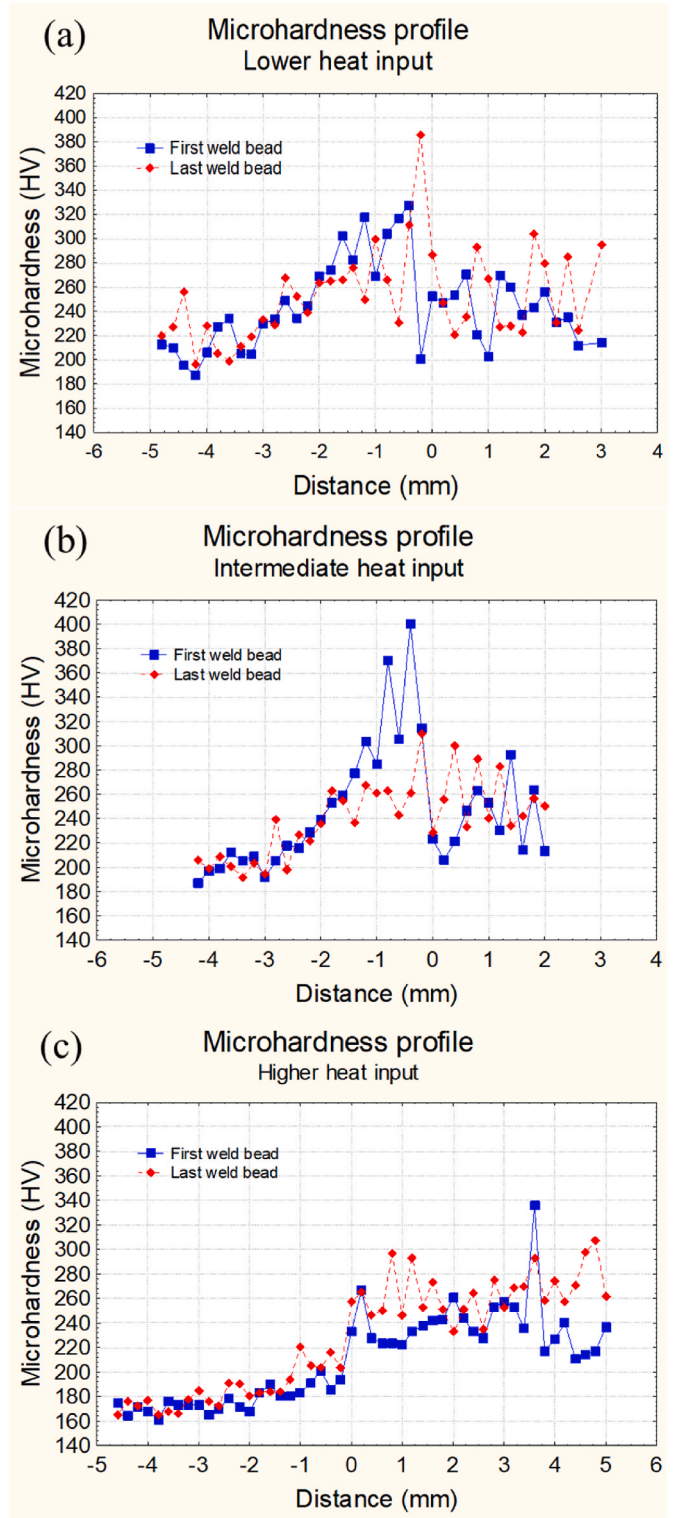


Fig. 15. Microhardness profile crossing the weld interface. (a) T1 sample; (b) T2 sample; (c) T3 sample.

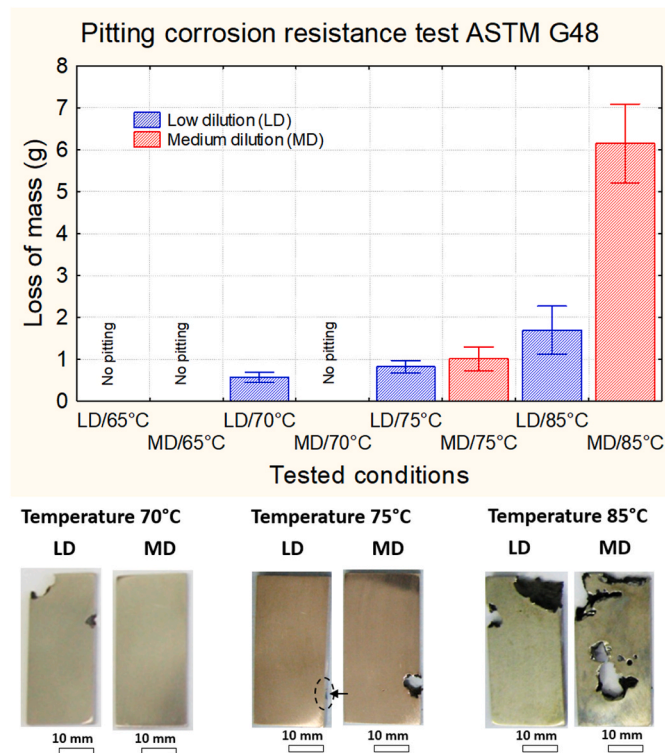


Fig. 16. Graphic and sample images of the pitting corrosion testing following the ASTM G48 standard.

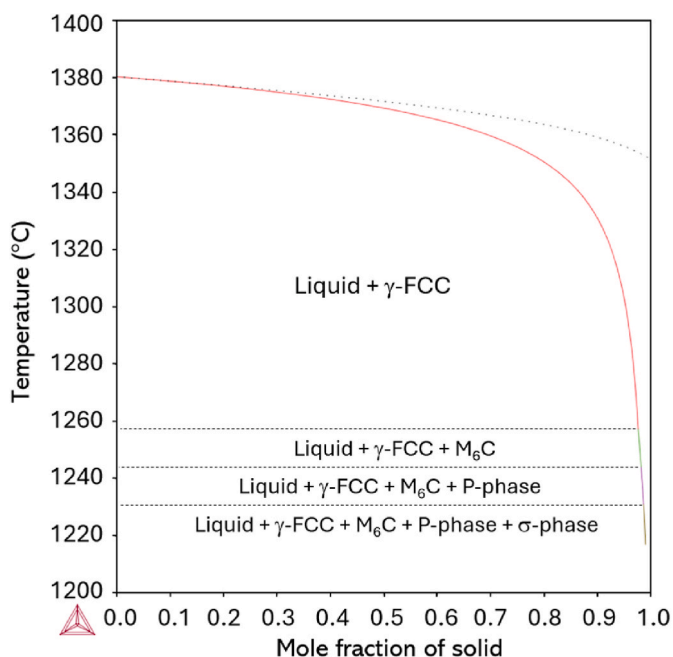


Fig. 17. Scheil-Gulliver diagrams (Solid fraction versus temperature) of Alloy C-276 diluted with carbon steel.

of the alloys was considerably more aggressive than in the low-dilution-welded samples. One reason for this may be secondary-phase precipitation.

Several factors influence the increase in the area fraction of the secondary phases. Higher heat input alters the solidification rate, affecting the microsegregation behaviour. However, as the heat input increases, the wire feed speed can also be increased, thereby reducing

the dilution level, as previously noted. Lower heat input results in higher dilution, thereby decreasing the alloying element content in the nickel-based alloy weld metal due to Fe introduced from the steel. This reduction in elements such as Mo, W, and Cr reduces the likelihood of segregation and secondary-phase formation. Still, it reduces the content of indispensable alloying elements crucial for corrosion resistance. On the other hand, higher heat input reduces dilution of the carbon steel substrate, whereas the slower solidification rate promotes the formation of larger and more numerous secondary phases. Therefore, the higher precipitate concentration in the low-dilution sample provides more nucleation sites for pits. Precipitates serve as potent cathodic sites, transforming the adjacent metal into a highly susceptible anodic region, thereby significantly influencing corrosion processes. In addition, some precipitates, such as the *P*-phase, exhibit pronounced Mo and W depletion in the nanometric vicinity of their surfaces [60]. This localised compositional change significantly alters their passivation and re-passivation capabilities, indicating a strong influence on their protective properties.

Conversely, the lower Fe content increased the availability of Cr, Mo, and W in solid solution, which may promote re-passivation after pitting. Thus, although the pit nucleates slightly earlier, possibly caused by the increased number of sites along the matrix-precipitate interface, it does not expand as much with increasing temperature. The medium-dilution condition, by contrast, had fewer precipitates, thereby hindering nucleation. However, once pitting initiates, it propagates more aggressively, likely due to increased Fe content and reduced availability of essential corrosion-resistant alloying elements, resulting in greater overall material loss.

5. Conclusion

Based on the experimental results obtained in this study concerning the dilution effect of the dissimilar weld overlays of Hastelloy C276 alloy with low carbon steel on the microsegregation behaviour and resultant microstructure produced by GTAW with an automatic wire feeding process, we can conclude the following:

- Increasing the heat input alongside the wire feed melting rate resulted in a decrease in the dilution level, which influenced the chemistry of the fusion zone. Additionally, the rise in heat input affects its microsegregation and, consequently, the volume fraction of secondary phases.
- Ni, Cr, Mo, Fe, and W exhibit distinct microsegregation behaviours: Mo strongly segregates to the interdendritic regions, while Fe and Ni concentrate in the dendrite cores. W showed no clear evidence of segregation towards the interdendritic areas. However, an increase in its concentration in secondary phases was observed. Cr does not segregate, maintaining a homogeneous distribution in both the core and interdendrite.
- The higher concentrations of Mo and W within the interdendritic regions strongly influenced the development of microstructural features in the dissimilar welds, as Mo-rich TCP phases were observed. The volume fraction of the Mo-rich secondary phases in the weld metal decreased with increasing dilution, due to the increase in the Fe content and, consequently, a decrease in the concentrations of the alloying elements.
- The microstructure of the dissimilar weld overlay deposited with Hastelloy C276 on low-carbon steel was formed by a γ -fcc Ni matrix with a dendritic pattern and secondary phases formed by topologically close-packed phases rich in Mo and W, such as the *P*-, σ - and μ -phases, in minor quantities.
- The significant alteration in chemical composition along the interface led to the formation of a high-dilution zone (HDZ). When this HDZ gradually changes its composition from the base to the weld metal, exhibiting a uniform chemical composition gradient, it is termed a continuous C-HDZ.

- If inhomogeneous distributions arise from the interaction of convection fronts, a heterogeneous pattern forms, resulting in Fe-rich zones intercalated with zones richer in filler metal at the interface, which consists of a martensitic microstructure with high hardness, called discontinuous D-HDZ.
- The critical pitting temperature (CPT) achieved by the diluted weld metals of Hastelloy C276 showed variations in its evolution, suggesting some effect of secondary phases precipitation on the nucleation pits, and the influence of dilution and Fe content on repassivation behaviour, due to differences in mass loss related to the temperature test.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Cleiton Carvalho Silva reports financial support was provided by Coordination for the improvement of Higher Education Personnel. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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